

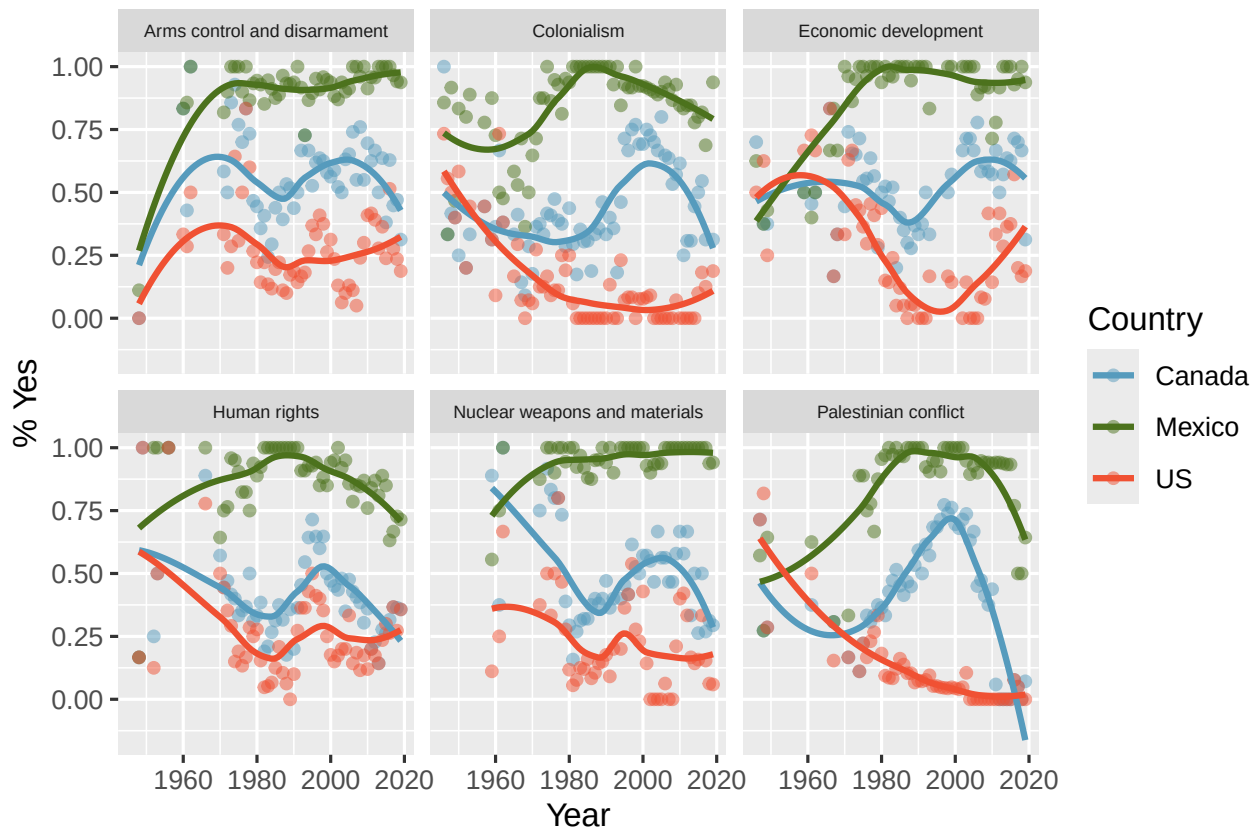
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1 Intro to data

Exercise 1.1. Within the framework of the Munich Rent Report, several variables were surveyed. Discuss the variables **net rent** (in Euro), **living size** (in m²), **building age** (in years), **central heating** (yes/no), **hot water supply** (yes/no), **good location** (yes/no), and **upscale bathroom equipment** (yes/no) regarding their variable type (i.e., nominal, ordinal, numeric).

Exercise 1.2. The visualization below shows voting patterns on various issues in the United States, Canada, and Mexico in the United Nations General Assembly. Precisely, for a given year between 1946 and 2015, it displays the percentage of roll calls in which the country voted yes for each issue. This visualization was constructed based on a dataset where each observation is a country/year pair.



- List the variables used in creating this visualization.
- Indicate whether each variable in the study is numerical or categorical. If categorical, indicate if the variable is ordinal.

Exercise 1.3. A statistical consulting company is charged with the following topics:

- a) Quality tests of wine from Koblenz.
- b) Quality inspections of condom production.
- c) Analyzing the addiction behavior of youths.

As head of the data collection department, you should decide between a census (full survey of the whole population) and a sample survey to collect the required data. Give reasons for your decisions.

Exercise 1.4. To improve the planning of parking spaces and the use of company-owned buses, a company is interested in some information about its employees. Specifically, the company is interested in the distance from their place of residence and the means of transport predominantly used by their employees to reach the workplace. To this end, the company uses a survey in which the economic situation of the employees was assessed. On a reference date, 50 employees were selected and questioned on the following aspects:

- household size (number of people living in the household),
- monthly rent,
- means of transport predominantly used to reach the workplace,
- distance between the place of residence and the place of work,
- own assessment of economic situation with 1 = very good, ... , 5 = very bad.

- a) Specify the population and the observational units (the cases).
- b) Which values can the collected variables take, and of which type are these variables?

Exercise 1.5. A study that surveyed a random sample of otherwise healthy high school students found that they are more likely to get muscle cramps when they are stressed. The study also noted that students drink more coffee and sleep less when they are stressed.

- a) What type of study is this?
- b) Can this study be used to conclude a causal relationship between increased stress and muscle cramps?
- c) State possible confounding variables that might explain the observed relationship between increased stress and muscle cramps.

Exercise 1.6. A researcher is interested in the effects of exercise on mental health and he proposes the following study:

- Use stratified random sampling to ensure representative proportions of individuals aged 18-30, 31-40, and 41-55 from the population.
- Next, randomly assign half the subjects from each age group to exercise twice a week, and instruct the rest not to exercise.
- Conduct a mental health exam at the beginning and at the end of the study, and compare the results.

- a) What type of study is this?
- b) What are the treatment and control groups in this study?
- c) Does this study make use of blocking? If so, what is the blocking variable?
- d) Does this study make use of blinding?
- e) Comment on whether or not the study results can be used to establish a causal relationship between exercise and mental health, and indicate whether or not the conclusions can be generalized to the population at large.
- f) Suppose you are tasked with determining if this proposed study should get funding. Would you have any reservations about the study proposal?

2 Data vizualization / wrangling

Remark: Use the Quarto file `starwars-template.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 2.1. In this exercise, we will use the `starwars` dataset, which is included in the `dplyr` package. Because `ggplot2` is also required, we will load the `tidyverse` package.

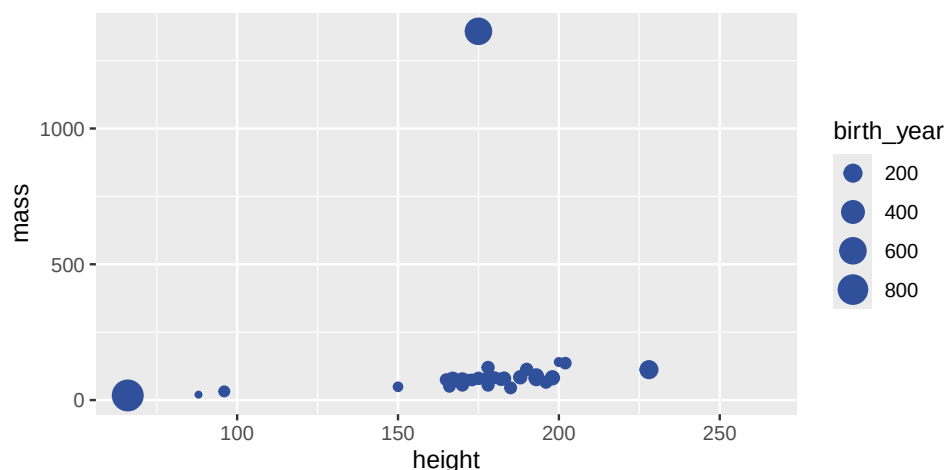
```
library(tidyverse)
```

Begin by examining the data. The `starwars` object is a tibble, so running the following code provides a concise overview of its structure.

```
starwars
# # A tibble: 87 x 14
#   name      height  mass hair_color skin_color eye_color birth_year sex  gender
#   <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
# 1 Luke Sk~    172    77 blond      fair        blue         19  male  mascu~
# 2 C-3P0      167    75 <NA>       gold        yellow       112  none  mascu~
# 3 R2-D2       96    32 <NA>       white, bl~  red         33  none  mascu~
# 4 Darth V~   202   136 none       white       yellow      41.9  male  mascu~
# 5 Leia Or~   150    49 brown      light       brown        19  fema~  femin~
# 6 Owen La~   178   120 brown, gr~ light       blue         52  male  mascu~
# 7 Beru Wh~   165    75 brown      light       blue         47  fema~  femin~
# 8 R5-D4       97    32 <NA>       white, red  red         NA  none  mascu~
# 9 Biggs D~   183    84 black      light       brown        24  male  mascu~
# 10 Obi-Wan~  182    77 auburn, w~ fair        blue-gray    57  male  mascu~
# # i 77 more rows
# # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
# #   vehicles <list>, starships <list>
```

a) Modify the following plot so that all points are displayed in pink.

```
ggplot(starwars,
       aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "#30509C")
```



b) Add labels for the plot title, x-axis, y-axis, and point size. Uncomment the relevant lines to observe the changes.


```
ggplot(starwars,
       aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "#30509C") +
  labs(
    #title = "___",
    #x = "___",
    #y = "___",
    #___
  )
```

- c) Select one categorical variable from the dataset and create a bar plot to display its distribution.

A bit of starter code is provided below, and the code chunk is set not to be evaluated with `eval: false` because the current code is not valid, so that the document would not render. Once you replace the code with valid code, set the chunk option to `eval: true` or remove the `eval` option altogether since it is set to `true` by default.

```
ggplot(starwars, aes(___)) +
  geom___
```

- d) Select one numerical variable and create a histogram to visualize its distribution.

This time, no starter code is provided. You are on your own!

```
# add code here
# pay attention to correctness and code style
```

- e) Select one numerical and one categorical variable, and create a visualization of their relationship using a method of your choice. Provide both your code and an interpretation of the resulting plot.

```
# add code here
# pay attention to correctness and code style
```

- f) Select two numerical and two categorical variables, and construct a visualization that incorporates all four variables. Include an interpretation with your response.

```
# add code here
# pay attention to correctness and code style
```

Remark: Use the Quarto file `hotels-template.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 2.2.

```
library(tidyverse)
```

The dataset used in this exercise comes from Tidy Tuesday. It's data about hotel bookings. Since the data is not contained in some package, we have to read in the data on our own. To do so, we use the `read_csv()` function from the `readr` (part of the `tidyverse`) package.

```
hotels <- read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidyuesday/master/data/2020/2020-02-11/hotels.csv"
```

Remark. The `read_csv()` function works not only if the data file is stored on your computer, you can also use it to read csv files that are available on a web server.

- a) Warm up! Take a look at an overview of the data with the `summary()` function.

Note

I already gave you the answer to this exercise. You only need to render the document and view the output. All variables are defined in the Data Dictionary, which is part of the template `hotels-template.qmd`. Thus, you don't need to familiarize yourself with all variables to work through these exercises.

```
summary(hotels)
```

- b) Are people traveling on a whim? Let's see...

Fill in the blanks for filtering for hotel bookings where the guest is **not** from the US (country code "USA") and the `lead_time` is less than one day.

Note

You will need to set `eval: true` when you have an answer you want to try out.

```
hotels |>
  filter(
    country ____ "USA",
    lead_time ____ ____
  )
```

- c) How many bookings involve at least one child **or** baby?

In the following chunk, replace

- [AT LEAST] with the logical operator for "at least" (in two places)
- [OR] with the logical operator for "or"

```
hotels |>
  filter(
    children [AT LEAST] 1 [OR] babies [AT LEAST] 1
  )
```

- d) Do you think it's more likely to find bookings with children or babies in city hotels or resort hotels? Test your intuition. Use `filter()` to determine the number of bookings in resort hotels with at least one child **or** baby in the room. Then, do the same for city hotels and compare the numbers of rows in the resulting filtered data frames.

```
# add code here
# pay attention to correctness and code style
```

```
# add code here
# pay attention to correctness and code style
```

- e) Create a frequency table of the number of `adults` in a booking. Display the results in descending order so the most common observation is on top. What is the most common number of adults in bookings in this dataset? Are there any surprising results?

Note

Don't forget to label your R chunk as well (where it says `label-me-1`). Your label should be short, informative, and shouldn't include spaces. It also shouldn't repeat a previous label, otherwise Quarto will give you an error about repeated R chunk labels.

```
# add code here
# pay attention to correctness and code style
```

- f) Repeat e) once for canceled bookings (`is_canceled` coded as 1) and once for not canceled bookings (`is_canceled` coded as 0). What does this reveal about the surprising results you spotted in the previous exercise?

```
# add code here
# pay attention to correctness and code style
```

3 Explore data

Exercise 3.1. Consider the American Community Survey (2012). Data from this survey is contained in the data set `acs12` from the `openintro` package.

- a) Use R to compute the mean and median of the variable `time to work` for each level of the variable `race`.
- b) What kind of shape has the distribution of `time_to_work` for each level of `race`? Answer this question by producing boxplots of `time_to_work` for each level of `race`.
- c) Analyse the association between travel time to work and income. Explain your findings.

Exercise 3.2. A company surveyed 910 randomly sampled customers at a large shopping mall about their preferred type of reusable water bottle. Respondents could choose:

- (i) Stainless steel
- (ii) Plastic (BPA-free)
- (iii) Glass
- (iv) Not sure

The results are shown by age group:

	<i>Age group</i>			Total
	18–29	30–49	50+	
<i>(i)</i> Stainless steel	84	112	82	278
<i>(ii)</i> Plastic (BPA-free)	103	118	41	262
<i>(iii)</i> Glass	132	98	120	350
<i>(iv)</i> Not sure	7	5	8	20
Total	326	333	251	910

- a) What percent of these shoppers are in the 18–29 age group?
- b) What percent of these shoppers prefer stainless steel bottles?
- c) What percent of these shoppers are 18–29 and prefer stainless steel?
- d) Among each age group, what percent prefers stainless steel?
- e) Do age group and bottle preference appear to be independent? Explain.

Remark: Use the Quarto file `starwars-template-2.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 3.3. We want to work with the `starwars` dataset, which is part of the `dplyr` package. Since we also need `ggplot2`, we load the `tidyverse`.

```
library(tidyverse)
```

- a) Visualize the distribution of the `height`, `mass` and `birth_year`. Describe the shape for each of the three distributions. Complete the following code.

```
ggplot(starwars, aes(x = height)) +  
  ---  
  
ggplot(starwars, aes(x = mass)) +  
  ---  
  
ggplot(starwars, aes(x = birth_year)) +  
  ---
```

- b) Visualize the relation between the variables `mass` and `height`, including the information about `hair_color` and `gender`. Provide an interpretation with your answer. Use the function `drop_na()` to remove observations that miss `gender` and `hair_color` information when creating the plot.

```
# add code here  
# pay attention to correctness and code style
```

- c) Calculate minimum, mean, median, and maximum height grouped by the actual sex of the character. Are men larger on average?

Hint: be aware that there are missing values in the data set.

```
# add code here  
# pay attention to correctness and code style
```

Exercise 3.4. Consider the following paired numerical data describing the weekly study hours (**hours**) and exam scores (**score**) of 12 students:

```
df <- tibble(  
  hours = c(6, 8, 3, 11, 7, 12, 9, 4, 10, 5, 6, 9),  
  score = c(59, 66, 45, 78, 63, 80, 75, 49, 72, 55, 63, 70),  
  student = 1:12  
)
```

- a) Create a scatterplot of **hours** (x-axis) and **score** (y-axis). Describe whether a *linear* relationship appears plausible.
- i) Compute the empirical correlation coefficient $r_{(x,y)} = \frac{s_{xy}}{s_x s_y}$ and interpret the result in terms of direction, strength, and linearity.

Note: Don't use R (the function `cor()`) to compute the correlation. Use instead the below given values and the formula defining the correlation coefficient.

```
sum((df$hours - mean(df$hours)) * (df$score - mean(df$score)))  
# [1] 337.5  
sd(df$hours)  
# [1] 2.812311  
sd(df$score)  
# [1] 11.14743
```

- b) Repeat the calculation after replacing the score of the 6th student with 55 (keeping **hours** = 12). How does $r_{(x,y)}$ change? What does this illustrate about the sensitivity of the Pearson correlation?

Hint: Create a new dataset `df_extreme` where you change the score of the 6th student. Then compute the correlation using `cor()`.

- c) Suppose the students belong to two study programs. Students 1 to 6 belong to program A and the rest to program B. Add a variable **program** to `df`, and then compute the correlation separately for programs A and B. Compare the groupwise correlations to the overall correlation and comment.

4 Probability

Exercise 4.1. Assortative mating is a non-random mating pattern where individuals with similar genotypes and/or phenotypes mate more frequently than expected under a random mating pattern. Researchers studying this topic collected data on the eye colors of 204 Scandinavian men and their female partners. The table below summarizes the results.

		<i>Partner (female)</i>			Total
		Blue	Brown	Green	
<i>Self (male)</i>	Blue	78	23	13	114
	Brown	19	23	12	54
	Green	11	9	16	36
	Total	108	55	41	204

- What is the probability that a randomly chosen male respondent or his partner has blue eyes?
- What is the probability that a randomly chosen male respondent with blue eyes has a partner with blue eyes?
- What is the probability that a randomly chosen male respondent with brown eyes has a partner with blue eyes? What is the probability of a randomly chosen male respondent with green eyes having a partner with blue eyes?
- Does the eye color of male respondents and their partners appear to be independent? Explain your reasoning.

Exercise 4.2. About 30% of human twins are identical, and the rest are fraternal. Identical twins are necessarily the same sex – half are males, and the other half are females. One-quarter of fraternal twins are both males, one-quarter are both females, and one-half are mixed: one male, one female. You have just become a parent of twins and are told they are both girls. Given this information, what is the probability that they are identical?

Exercise 4.3. A genetic test is used to determine if people have a predisposition for *thrombosis*, which is the formation of a blood clot inside a blood vessel that obstructs blood flow through the circulatory system. It is believed that 3% of people have this predisposition. The genetic test is 99% accurate if a person has the predisposition, meaning that the probability of a positive test result when a person actually has the predisposition is 0.99. The test is 98% accurate if a person does not have the predisposition. What is the probability that a randomly selected person who tests positive for the predisposition by the test actually has the predisposition?

Exercise 4.4. Consider the following card game with a well-shuffled standard 52-card deck. If you draw a red card, you win nothing. If you get a spade, you win \$5. For any club, you win \$10 plus an extra \$20 for the ace of clubs.

- a) Create a probability model for the amount you win at this game. Also, find the expected winnings for a single game and the standard deviation of the winnings.
- b) What is the maximum amount you would be willing to pay to play this game? Explain your reasoning.

Exercise 4.5. An airline charges the following baggage fees: \$25 for the first bag and \$35 for the second. Suppose 54% of passengers have no checked baggage, 34% have one piece of checked baggage, and 12% have two pieces. We suppose a negligible portion of people check more than two bags.

- a) Build a probability model, compute the average revenue per passenger, and compute the corresponding standard deviation.
- b) About how much revenue should the airline expect for a flight of 120 passengers? With what standard deviation? Note any assumptions you make and if you think they are justified.

Exercise 4.6. During triathlons, participants are usually grouped by age and gender. We want to examine the time it takes to finish the triathlon. A suitable probability model for triathlon times is the normal distribution. However, each age and gender group will have its own distinct normal distribution. We will focus on the following two groups:

- Men, Ages 30-34: $\mathcal{N}(4313, 583^2)$
- Women, Ages 25-29: $\mathcal{N}(5261, 807^2)$

Times are listed in seconds. Use this information to compute each of the following:

- a) The cutoff time for the fastest 5% of athletes in the men's group, i.e., those who took the shortest 5% of time to finish.
- b) The cutoff time for the slowest 10% of athletes in the women's group.

Hint: Use R to compute any probability or quantile in this exercise.

Exercise 4.7. The Capital Asset Pricing Model (CAPM) is a financial model that assumes returns on a portfolio are normally distributed. Let's consider a portfolio with an average annual return of 14.7% (i.e., an average gain of 14.7%) and a standard deviation of 33%. A 0% return indicates that the portfolio's value remains unchanged, a negative return signifies a loss, and a positive return indicates a gain.

- a) What percent of years does this portfolio lose money, i.e. have a return less than 0%?
- b) What is the cutoff for the highest 15% of annual returns with this portfolio?

Hint: Use R to compute any probability or quantile in this exercise.

5 Predictive modelling

Exercise 5.1. For each part a) through d), indicate whether we would generally expect the performance of a non-parametric statistical learning method to be better or worse than a parametric method. Justify your answer.

- a) The sample size n is extremely large, and the number of predictors p is small.
- b) The number of predictors p is extremely large, and the number of observations n is small.
- c) The relationship between the predictors and response is highly non-linear.
- d) The variance of the error terms, i.e., $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.

Exercise 5.2. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in predicting an outcome variable or understanding the relation between the input variables and the output variable. Finally, provide the number n of distinct data points and the number of variables p that are available for use in making predictions.

- a) We collect data on the top 500 firms in the US. We record each firm's profit, number of employees, industry, and CEO salary. We are interested in understanding which factors affect CEO salary.
- b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product, we have recorded whether it was a success or a failure, the price charged, the marketing budget, the competitor's price, and 10 other variables.
- c) We are interested in predicting the % change in the USD/Euro exchange rate in relation to weekly changes in global stock markets. Hence, we collected weekly data for the entire year of 2012. For each week, we recorded the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Exercise 5.3. We will now think of some real-life applications for different statistical learning methods.

- a) Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference (describing the relationship) or prediction? Explain your answer.
- b) Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference (describing the relationship) or prediction? Explain your answer.

Exercise 5.4. The following training data, containing six observations on three predictors and one qualitative response variable, is given.

```
train
#   x1 x2 x3   y
# 1  0  3  0  red
# 2  2  0  0  red
# 3  0  1  3  red
# 4  0  1  2 green
# 5 -1  0  1 green
# 6  1  1  1  red
```

Suppose we wish to use this dataset to predict a value for the response Y given the new observation:

```
new
#   x1 x2 x3
# 1  0  0  0
```

using k-nearest neighbors.

- Compute the euclidean distance between each observation and the new test point, **new**.
- What is your prediction for $k = 1$? Justify your answer.
- What is our prediction for $k = 3$? Justify your answer.

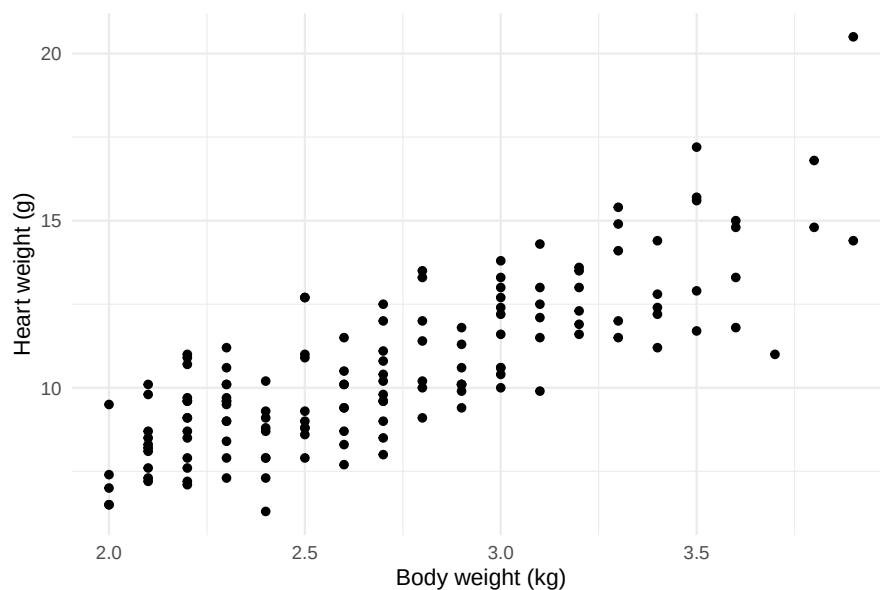
Exercise 5.5. Data are provided on the shoulder girth and height of a group of individuals. The mean shoulder girth is 107.20 cm (standard deviation: 10.37 cm), and the mean height is 171.14 cm (standard deviation: 9.41 cm). The correlation coefficient between height and shoulder girth is 0.67.

- a) Write the equation of the fitted regression model for predicting height.
- b) Interpret the slope and the intercept in this context.
- c) Calculate R^2 of the regression line for predicting height from shoulder girth, and interpret it in the context of the application.
- d) A randomly selected student from your class has a shoulder girth of 100 cm. Predict the student's height using the model.
- e) The student from d) is 160 cm tall. Calculate the residual and explain what this residual means.
- f) A one-year-old has a shoulder girth of 56 cm. Would it be appropriate to use this linear model to predict the height of this child?

Exercise 5.6. The following regression model

```
mod <- lm(Hwt ~ Bwt, data = cats)
```

is used for predicting the heart weight **Hwt** (in g) of cats from their body weight **Bwt** (in kg).



The coefficients are estimated using a dataset of 144 domestic cats. One obtains the following least squares estimates

```
coef(mod)
# (Intercept)      Bwt
#      -0.357      4.034
```

- Write out the fitted linear model.
- Interpret the intercept.
- Interpret the slope.
- Identify the R^2 value in

```
broom::glance(mod)
# # A tibble: 1 x 12
#   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC    BIC
#   <dbl>      <dbl> <dbl>     <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
# 1    0.647        0.644  1.45      260. 6.97e-34     1  -257.  520.  529.
# # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

and give an interpretation of the computed R^2 value.

- Calculate the empirical correlation coefficient between the response and the predictor variable.

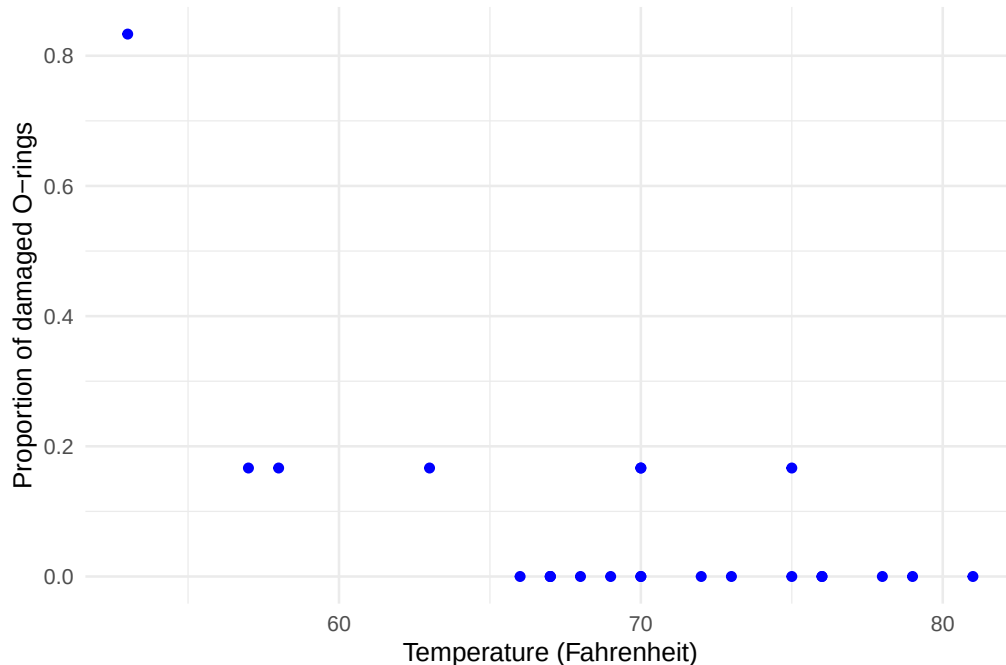
Exercise 5.7. In this exercise, we are using a forward-selection algorithm, where we attempt to add variables to the model one at a time. The table below displays the initial step of the forward-selection process, beginning with an intercept model and considering the variables `ethnicity`, `gender`, and `cls_students` within the scope. Based on this table, which variable should be added to the model in the first step, and why?

```
Start:  AIC=-562.99
score ~ 1
```

	Df	Sum of Sq	RSS	AIC
+ cls_students	1	0.09216	136.56	-561.30
+ ethnicity	1	0.78567	135.87	-563.66
+ gender	1	2.26021	134.39	-568.71
<none>			136.65	-562.99

Hint: The table is not given in standard form.

Exercise 5.8. On January 28, 1986, the Challenger space shuttle was scheduled for a routine launch. Seventy-three seconds after liftoff, the shuttle disintegrated, causing the deaths of all seven crew members. The investigation identified a critical seal, known as an O-ring, as a primary factor in the disaster. Analysis revealed a strong association between ambient temperature at launch and O-ring damage. The following plot displays the proportion of damaged O-rings at each recorded temperature. Data on O-ring performance were collected from 23 shuttle missions, with ambient temperature recorded for each launch. Six O-rings were tested per mission.



- a) The logistic model fit to these data may be written as

$$\log \left(\frac{\hat{p}(x)}{1 - \hat{p}(x)} \right) = 11.6630 - 0.2162 \times x$$

where $\hat{p}$ is the model-estimated probability that an O-ring will become damaged and x is the temperature in Fahrenheit. Use the model to calculate the probability of O-ring damage at ambient temperatures of 51, 53, and 55 degrees Fahrenheit. The model-estimated probabilities for various additional ambient temperatures are provided below.

$\hat{p}(57) = 0.341$	$\hat{p}(59) = 0.251$	$\hat{p}(61) = 0.179$	$\hat{p}(63) = 0.124$
$\hat{p}(65) = 0.084$	$\hat{p}(67) = 0.056$	$\hat{p}(69) = 0.037$	$\hat{p}(71) = 0.024$

- b) Add the model-estimated probabilities from a) on the plot, then connect these dots using a smooth curve to represent the model-estimated probabilities.
Hint: If you want to add the curve using R, you can do so. The dataset is called `openintro::orings`.
- c) Describe any concerns you may have regarding applying logistic regression in this application, and note any assumptions that are required to accept the model's validity.

Exercise 5.9. Determine which of the following statements are true and which are false. For each false statement, provide an explanation for why it is incorrect.

- a) Suppose we have the first two observations from a logistic regression model. In observation 1, the first variable takes the value $x_1 = 6$, and in observation 2, it takes the value $x_1 = 4$. Suppose errors were made in recording the two observations. The correct value for the first observation is $x_1 = 7$ (instead of 6), and for the second observation, $x_1 = 5$ (instead of 4). After correcting these values, the predicted probability from the logistic regression model would increase by the same amount for each observation.
- b) When using a logistic regression model, the model cannot predict a probability that is negative or greater than 1.
- c) Because logistic regression predicts probabilities of outcomes, the observations used to build a logistic regression model need not be independent.
- d) When we fit a logistic regression model, we usually perform model selection using adjusted R^2 .

6 Foundations of Inference

Exercise 6.1. A recent survey of 840 adults in Germany found that 268 respondents said they would be unable to cover an unexpected 5000-Euro expense without borrowing money or going into debt.

- a) What population is under consideration in the dataset?
- b) What parameter is being estimated?
- c) What is the point estimate for the parameter?
- d) Use the function `rep()` and `data.frame()` to create the sample data in R.

Hint: Create a data frame with one column named `expense`, containing 840 entries. 572 entries should be `"cover"` and the remaining 268 entries `"not cover"`.

- e) Use functions from the `infer` package to create a bootstrap distribution for the point estimate from part c) based on 1000 resamples.
- f) Compute the standard deviation of the point estimator based on the 1000 point estimates computed in part e).
- g) A politician claims that 40% of German adults cannot cover a 5000-Euro emergency expense. Based on the bootstrap distribution, should she be surprised by the data?

Exercise 6.2. A national poll conducted in 2013 found that 52% of U.S. adult Twitter users get at least some news on Twitter, and the standard error for this estimate was 2.4%. Identify each of the following statements as true or false. Provide an explanation to justify each of your answers.

- a) The data provide evidence with a confidence level of $1 - \alpha = 0.99$, indicating that more than half of U.S. adult Twitter users receive some news through Twitter.

Remark: This is a conservative statement.

- b) Since the standard error is 2.4%, we can conclude that 97.6% of all U.S. adult Twitter users were included in the study.
- c) If we want to reduce the standard error of the estimate, we should collect less data.
- d) If we construct a 90% confidence interval for the percentage of U.S. adults Twitter users who get some news through Twitter, this confidence interval will be wider than a corresponding 99% confidence interval.

Exercise 6.3. We are provided with a 95% confidence interval for the mean waiting time at an emergency room (ER) of (128, 147) minutes. Answer the following questions based on this interval.

- a) A local newspaper claims that the average waiting time at this ER exceeds 3 hours. Is this claim supported by the confidence interval? Explain your reasoning.
- b) The Dean of Medicine at this hospital claims the average wait time is 2.2 hours. Is this claim supported by the confidence interval? Explain your reasoning.
- c) Without actually calculating the interval, determine if the claim of the Dean from b) would be supported based on a 99% confidence interval?

Exercise 6.4. State the null and alternative hypotheses in both words and symbols for each scenario described below.

- a) In Florence, Italy, a new regulation requires gelaterie (ice-cream shops) to display the calorie content of every scoop of gelato.

Before this rule, the average caloric intake per customer at a popular gelato shop was 950 calories.

After calorie posting began, a nutrition researcher collected a random sample of customers to see whether calorie labeling changed how much people eat.

She believes there is a difference in the average number of calories consumed.

- b) The government of the Faroe Islands wants to understand last year's alcohol consumption among adults.

The Nordic regional average is that 76% of adults consumed alcohol at least once in the past year.

To investigate whether the Faroese population differs from this regional value, officials surveyed a random sample of 620 adults regarding their alcohol consumption.

Exercise 6.5. A study suggests that 25% of 25-year-olds are married. However, you believe this is incorrect and decide to collect your own sample for a hypothesis test. In a random sample of 25-year-olds from census data, consisting of 776 individuals, it was found that 24% of them are married. A friend of yours offered to help you set up the hypothesis test and came up with the following hypotheses. Indicate any errors you see.

$$H_0 : \hat{\theta} = 0.24 \quad \text{vs.} \quad H_A : \hat{\theta} \neq 0.24.$$

Exercise 6.6. A chocolate producer suspects irregularities in her production. For quality control, she randomly selects 15 bars and measures their weight in grams. The variable $\mathbf{x}$ contains the sample observations. The producer has identified an issue with the average weight, which should be 100g. You can assume that the weight follows a normal distribution, meaning that $\mathbf{x}$ contains realizations of 15 i.i.d. random variables, denoted as $X_1, \dots, X_{15}$. Each X_i follows a normal distribution with mean μ and variance σ^2 , with both being unknown.

- Formulate null hypothesis and alternative hypothesis for a two-sided test for the average weight μ .
- Test the null hypothesis from part a) at the significance level $\alpha = 0.05$. Justify your answer using the hint provided.

Hint:

```
c(sum(x), sd(x))
```

```
# [1] 1482.07    1.69
```

```
pt(seq(-3, -2, by = 0.1), df = 14)
```

```
# [1] 0.00477576 0.00582103 0.00709016 0.00862833 0.01048867 0.01273333
```

```
# [7] 0.01543446 0.01867514 0.02255015 0.02716657 0.03264398
```

Exercise 6.7. Re-examine the data set provided in the file `gifted.csv`, available on Moodle.

Remember: Researchers collected a simple random sample of 36 children identified as gifted in a large city. The variables included in the data set are as follows:

- **score** Score in test of analytical skills.
- **fatheriq** Father's IQ.
- **motheriq** Mother's IQ.
- **speak** Age in months when the child first said 'mummy' or 'daddy'.
- **count** Age in months when the child first counted to 10 successfully.
- **read** Average number of hours per week the child's mother or father reads to the child.
- **edutv** Average number of hours per week the child watched an educational program on TV during the past three months.
- **cartoons** Average number of hours per week the child watched cartoons on TV during the past three months.

The researchers believe that gifted children spend more time watching cartoons than educational programs.

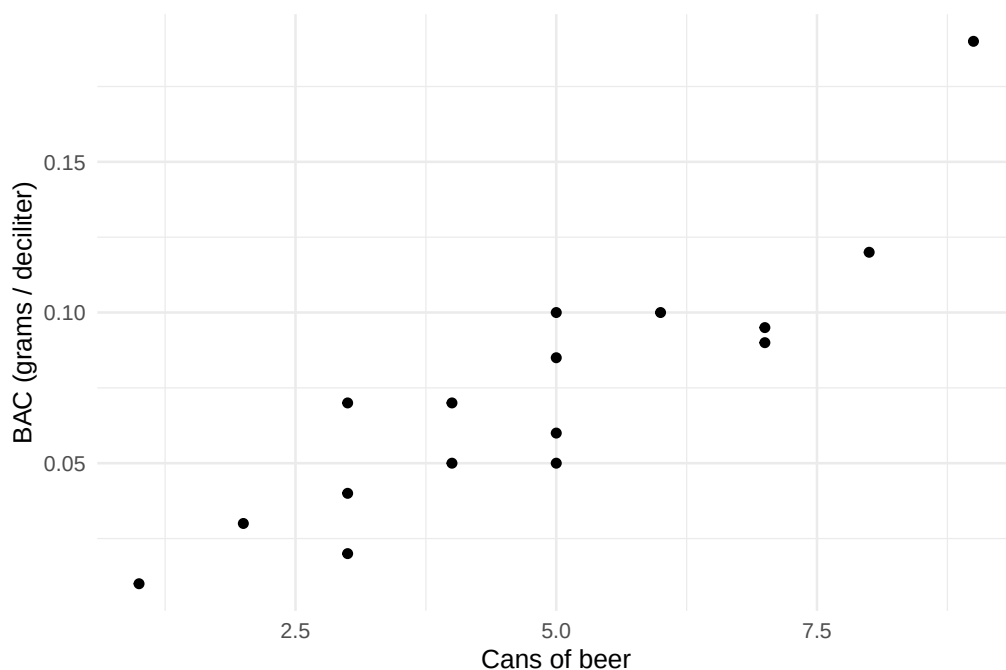
- a) State the null and alternative hypotheses in order to test the researchers' assumption.
- b) Use the simulation-based approach for testing the null hypothesis from part a). Decide if you can reject the null hypothesis at a significance level of $\alpha = 0.05$. State your conclusion in the context of the data.

7 Inference for linear regression

Exercise 7.1. Many people believe that gender, weight, drinking habits, and many other factors are much more important in predicting blood alcohol content (BAC) than simply considering the number of drinks a person consumed. Here, we examine data from 16 student volunteers recruited at a German university who each drank a randomly assigned number of cans of beer. These students were evenly divided between men and women, and they differed in weight and drinking habits. Thirty minutes later, a police officer measured their blood alcohol content (BAC) in grams of alcohol per deciliter of blood.

The following scatterplot visualizes the observed data and the R output summarizes the results of the fitted regression model.

```
# # A tibble: 2 x 5
#   term      estimate std.error statistic    p.value
#   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
# 1 (Intercept) -0.0127   0.0126    -1.00  0.332
# 2 beers       0.0180   0.00240     7.48 0.00000297
```



- Describe the relationship between the number of cans of beer and BAC.
- Write the equation of the fitted regression line. Interpret the slope and intercept in the context of the data.
- Do the data provide strong evidence that drinking more cans of beer is associated with an increase in blood alcohol? State the null and alternative hypotheses, report the p-value, and state your conclusion.
- The correlation coefficient for the number of cans of beer and BAC is 0.89. Calculate R^2 and interpret it in context.
- Suppose we visit a bar, ask people how many drinks they have had, and also take their BAC. Do you think the relationship between the number of drinks and BAC would be as strong as the relationship found in the study?

Exercise 7.2. We are given the regression output from a model for predicting the heart weight (in g) of cats from their body weight (in kg). The coefficients are estimated using a dataset of 144 domestic cats. The model output is also provided below.

```
# # A tibble: 2 x 5
#   term      estimate std.error statistic    p.value
#   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
# 1 (Intercept) -0.3567    0.6923   -0.5152 6.072e- 1
# 2 Bwt         4.034     0.2503   16.12   6.969e-34
```

- We see that the point estimate for the slope is positive. What are the hypotheses for evaluating whether body weight is positively associated with heart weight in the population of domestic cats?
- Test the hypotheses from part a) using a 5% significance level, and then report the conclusion of the test in the context of the data.
- Calculate a 95% confidence interval for the slope of body weight, and interpret it in the context of the data.

Hint:

```
qt(c(0.9, 0.95, 0.975), df = 142)
# [1] 1.29 1.66 1.98
```

- Do your results from the hypothesis test and the confidence interval agree? Explain.

Exercise 7.3. The data set `Cars93` contains 93 observations on motor cars sold in the US in 1993. To explain the logarithm of the variable `MPG.city` (City MPG - miles per US gallon by EPA rating), a linear model was fitted using the explanatory variables

- **EngineSize:** Engine size (liters)
- **Weight:** Weight (pounds)
- **Passengers:** Passenger capacity (persons)
- **Price:** Mid-range Price (in \$1000): average of `Min.Price` and `Max.Price`.

The summary of this model contains some missing values (NA).

```
summary(model)

#
# Call:
# lm(formula = log(MPG.city) ~ ., data = cars)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -0.2631 -0.0619  0.0058  0.0460  0.4169
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  4.06e+00   7.24e-02     NA      NA
# EngineSize  -3.80e-03   2.03e-02     NA      NA
# Weight       -3.14e-04   4.63e-05     NA      NA
# Passengers   6.31e-03   1.47e-02     NA      NA
# Price        -2.03e-03   1.70e-03     NA      NA
#
# Residual standard error: 0.106 on 88 degrees of freedom
# Multiple R-squared:  0.787,    Adjusted R-squared:  0.778
# F-statistic: 81.4 on 4 and 88 DF,  p-value: <2e-16
```

- Identify all variables, which are significant at the level $\alpha = 0.05$. Justify your answer.
- Give an interpretation of the slope of all variables that have a significant effect due to a).
- Now a second model was fitted:

```
model_small <- lm(log(MPG.city) ~ Weight + Price, data = Cars93)
```

which has an adjusted R^2 of 0.781858. Which model would you prefer based on this information? Justify your answer.

Hint:

```
qt(0.025, df = 88, lower.tail = FALSE)
# [1] 1.98729
```

Exercise 7.4. In 41 US cities, sulphur dioxide concentrations in the air were analysed with respect to several climatic and geographical variables. A multiple linear regression model was fitted using the three explanatory variables `temp` (average annual temperature in degrees F), `man_fac` (number of manufacturers employing >20 workers), and `wind` (average annual wind speed in miles per hour). The dependent variable was the logarithmic annual average sulphur dioxide concentration in micrograms per square meter `log(sd_qm)`.

The following summary (with some missing values) was obtained.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.4892948	0.9927641	7.544	5.41e-09	***
temp	-0.0556772	0.0116911	-4.762	2.93e-05	***
num_manfac	0.0005633	0.0001446	3.894	0.000398	***
wind	-0.1580268	0.0597864	-2.643	0.011972	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4972 on 37 degrees of freedom

Multiple R-squared: 0.5364, Adjusted R-squared: NA

F-statistic: 14.27 on 3 and 37 DF, p-value: 2.473e-06

- State the equation of the fitted regression model.
- Formulate the null and alternative hypotheses to test if `wind` has a significant effect on the sulphur dioxide concentration. Decide if it has a significant effect at level $\alpha = 0.05$.
- Compute the adjusted R^2 value.
- Why did it make sense to consider the logarithmic pollutant concentrations?

Hint:

```
var(log(sd_qm))
# [1] 0.493223
```